

Forder — Book IV

Proposition 12

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

For points A, B, C on a circle with centre O , if O and C are on the same side of AB , the angle ACB is half the central angle AOB .

1 Introduction

Forder IV.12 is the central inscribed-angle theorem in the circle block used by the present development. In Forder’s notation, if A, B, C lie on a circle with centre O , and O, C lie on the same side of the chord AB , then the angle ACB is one half of the central angle AOB [1].

The theorem looks elementary when stated in the language of angle measure. The formal reconstruction shows why the synthetic statement is substantially richer. There is no numerical angle measure and there is no operation “divide an angle by two”. Instead, “half” is represented by an actual interior bisector ray. The production theorem therefore constructs a point T such that

$$KT \text{ is interior to } \angle AKB, \quad \angle AKT \cong \angle BKT, \quad \angle ACB \cong \angle AKT.$$

The proof also has to make explicit the order distinctions that Forder compresses into a conventional diagram. After extending CK through the centre to an antipodal point X , the line KX becomes the reference axis. The proof then separates the following configurations:

A, K, C collinear,	diameter endpoint case,
B, K, C collinear,	symmetric diameter endpoint case,
A, B on opposite sides of KX ,	addition case,
A, B on the same side of KX ,	subtraction case.

This is the first proposition in the present Forder Book IV documentation where the difference between a short source proof and a complete synthetic formal reconstruction becomes especially visible.

The production module is

```
CGJteamLab/Proposition4_12.lean
```

and it exports both IV.12 and the corollary IV.12.1.

2 Classical Configuration

C and K on the same side of chord AB

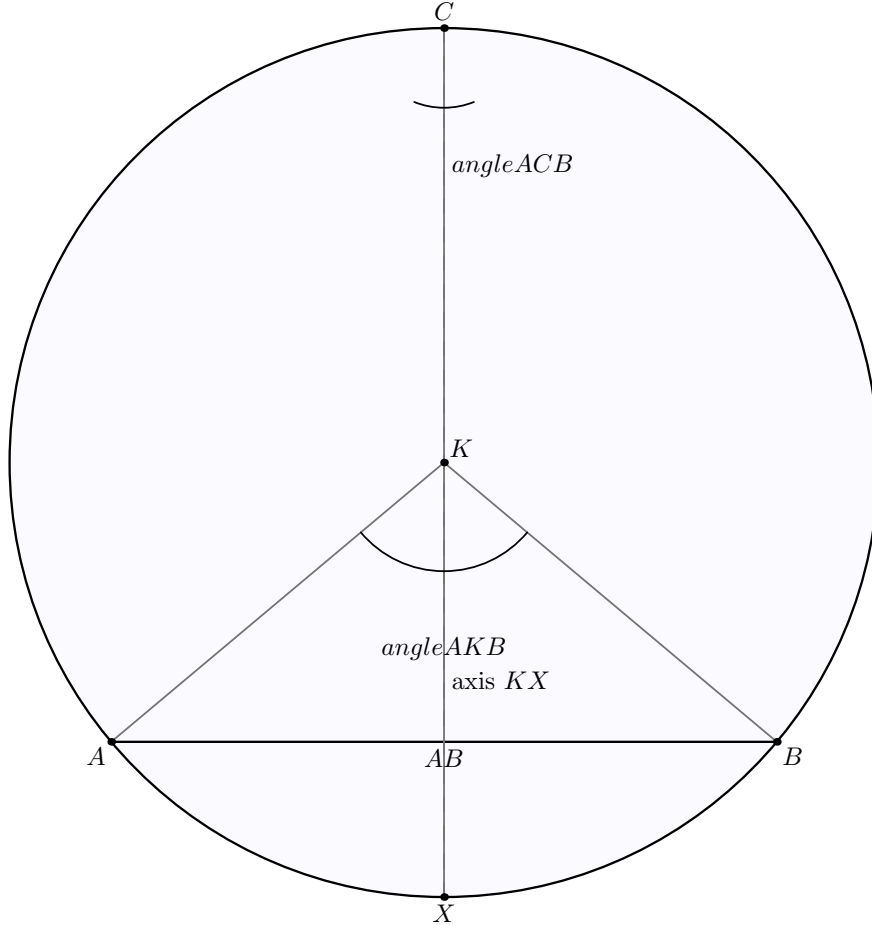


Figure 1: Global source configuration for Forder IV.12. The chord AB lies below the centre, so C and K are on the same side of the chord. The auxiliary antipodal point X determines the axis KX used in the formal case split.

Let K be the centre of a circle and let A, B, C be points on the circle. Let AB be the chord under consideration and assume that C and K lie in the same half-plane determined by the line AB .

The geometric content of IV.12 is

$$\angle ACB = \frac{1}{2} \angle AKB.$$

For the formal reconstruction this formula is only mnemonic. The actual synthetic target is the existence of an interior ray KT such that

$$\angle AKT \cong \angle TKB$$

and

$$\angle ACB \cong \angle AKT.$$

Extend CK through K to the antipodal point X , so that

$$C - K - X$$

and X lies on the same circle. The line KX is the axis used to organize the proof. Relative to this axis there are two generic configurations and two exceptional ones.

Addition configuration. If A and B lie on opposite sides of the axis KX , the axis passes through the central angle $\angle AKB$. The two exterior-central half-angle constructions attached to A, C, X and B, C, X contribute two pairs of equal angular components. The formal proof rearranges these four components to obtain a bisector of the whole central angle.

Subtraction configuration. If A and B lie on the same side of KX , one of the radius rays is nested inside the larger exterior-central angle determined by the other. Forder's subtraction of angle magnitudes must then be replaced by a synthetic comparison of two decompositions.

Diameter endpoint configurations. If A or B lies on the axis KX , then the corresponding chord through C passes through the centre and is therefore a diameter. The formal proof treats these configurations separately rather than forcing them through a generic angle-decomposition lemma.

3 Forder's Statement and Proof

Forder introduces IV.12 after explicitly saying that he wishes to treat angles in the same segment without invoking the full theory of angular measure, in particular avoiding reflex angles [1]. His statement is:

Let A, B, C be on a circle centre O , and let O, C be on the same side of AB ; then ACB is half AOB .

The proof extends CO through O to a point X on the circle and then separates the order configurations.

3.1 The diameter-aligned case

When X lies on OA (and symmetrically when it lies on OB), the line through C, O, X is also the diameter through A . Equal radii give an isosceles triangle on the remaining circle point. The central angle is then decomposed into two equal base-angle contributions, yielding the doubling relation required for IV.12.

This is the source ancestor of the two explicit production lemmas

```
hilbert_forder_IV12_diameter_A
hilbert_forder_IV12_diameter_B
```

3.2 The addition case

In the generic case Forder first obtains

$$\angle AOX = 2\angle OCA, \quad \angle BOX = 2\angle OCB.$$

If A and B lie on opposite sides of OX , then the ray OX lies inside the central angle and the corresponding ray through C lies inside the inscribed angle. Consequently the two decompositions add:

$$\begin{aligned}\angle AOB &= \angle AOX + \angle XOB, \\ \angle ACB &= \angle ACX + \angle XCB.\end{aligned}$$

Substituting the two exterior-central doubling relations gives

$$\angle AOB = 2\angle ACB.$$

In Lean this numerical-looking addition is replaced by an explicit rearrangement of interior rays and congruent angle components.

3.3 The subtraction case

If A and B lie on the same side of OX , one of the rays OA, OB lies inside the angle formed by the other ray and OX . Forder then writes the central and inscribed angles as differences:

$$\begin{aligned}\angle AOB &= \angle BOX - \angle AOX, \\ \angle ACB &= \angle OCB - \angle OCA\end{aligned}$$

up to the symmetric orientation. From the two doubled exterior-central relations he concludes again

$$\angle AOB = 2\angle ACB.$$

The formal proof cannot simply subtract angle measures. It reconstructs this step from nested rays, ordinary angle bisectors, uniqueness of halves, and the project's synthetic angle-decomposition theorems.

4 Formal Statement

The proposition interface is defined in `HilbertInterfaceIV.lean`:

```
def HilbertForderIV12CentralHalf
  [H : HilbertIncidence Geo]
  [_HO : @HilbertOrder Geo H] : Prop :=
  forall K R A B C : Geo.Point,
    forall chord : Geo.Line,
      Ne A B ->
        H.OnLine A chord ->
        H.OnLine B chord ->
        HilbertCircle Geo K R A ->
        HilbertCircle Geo K R B ->
        HilbertCircle Geo K R C ->
        HilbertSameSide Geo C K chord ->
        exists T : Geo.Point,
```

```
HilbertRayMeetsSegment Geo K T A B /\
Geo.AngleCongruent A K T B K T /\
Geo.AngleCongruent A C B A K T
```

The public production theorem is

```
theorem hilbert_IV12_central_half
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV12CentralHalf (Geo := Geo)
```

The distinction between the interface predicate and the theorem is useful. The predicate records the mathematical contract of IV.12 using only the order vocabulary needed to state it. The theorem then proves that contract under the project’s Euclidean-plane structure.

5 Why “Half an Angle” Is a Witnessed Construction

The conclusion

$$\angle ACB = \frac{1}{2} \angle AKB$$

cannot be copied literally into the synthetic language without introducing an arithmetic of angle magnitudes. The reconstruction instead asks for a point T satisfying three independent geometric obligations.

First,

```
HilbertRayMeetsSegment Geo K T A B
```

asserts that the ray KT genuinely lies inside the angle AKB . This is stronger than merely saying that T is noncollinear with the sides: the order witness records the interior position.

Second,

```
Geo.AngleCongruent A K T B K T
```

says that KT bisects the central angle.

Third,

```
Geo.AngleCongruent A C B A K T
```

identifies the inscribed angle with one of those two equal halves.

Thus IV.12 is formalized not as an equation of numbers but as the existence of a geometric configuration realizing the equation.

6 The Antipodal Reduction

The complete production proof begins by constructing the point X opposite C on the same diameter:

$$C - K - X.$$

The reusable theorem responsible for this construction is now

```
hilbert_circle_antipode
```

in `HilbertInterfaceIV`.

The proof then constructs the carrier line of KX and proves that C lies on it. This line is the axis relative to which all later order cases are classified.

Structurally the top-level proof has the form

```
by_cases hAaxis : H.OnLine A axis
...
by_cases hBaxis : H.OnLine B axis
...
by_cases hSameAB : HilbertSameSide Geo A B axis
...
```

The four outcomes are therefore exhaustive by incidence and plane separation; the final theorem introduces no additional geometric idea after the branch lemmas have been proved.

7 The Diameter Endpoint Branches

Suppose first that A, K, C are collinear. Because A and C lie on the circle and K is the centre, the reusable theorem

```
hilbert_circle_center_midpoint_of_chord
```

shows that K is the midpoint of AC . Hence

$$A - K - C.$$

Now consider triangle BCK . The radii KB and KC are congruent, so the triangle is isosceles and its base angles are congruent. Euclid I.32 is applied with CK produced through K to A . The exterior angle at K is thereby decomposed into two angles congruent to the equal base angles. This produces an actual bisector ray KP of the required central angle.

The final task is order-theoretic: the point P supplied by I.32 lies between B and A , so the ray KP is an interior ray of $\angle AKB$. Since C, K, A lie on one ray from C , angle transport identifies the original inscribed angle ACB with the appropriate base angle. The desired half-angle witness follows.

The B, K, C case is obtained by the symmetric production helper

case A, K, C collinear

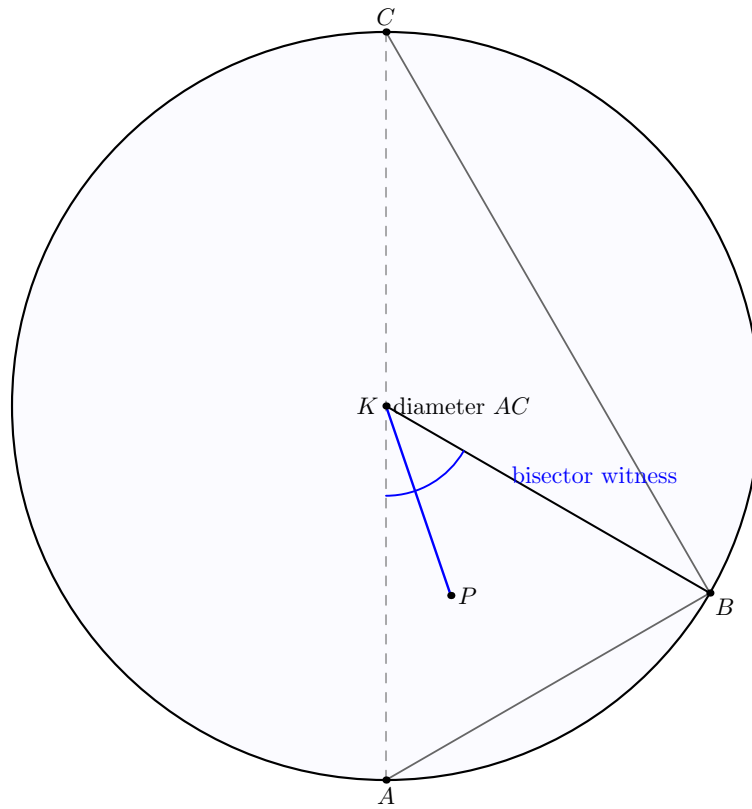


Figure 2: One of the two exceptional diameter branches: A, K, C are collinear, so AC is a diameter. The proof then uses the midpoint property of the centre together with Euclid I.32 to construct the required bisector witness.

```
hilbert_forder_IV12_diameter_B
```

The diameter branch is a useful indication of the proof's foundational profile: a source statement about circles reaches Euclid I.32 once the exceptional order configuration is made explicit.

8 The Exterior-Central Half Construction

Both generic branches use a reusable theorem extracted from the IV.12 development:

```
hilbert_circle_exterior_central_half
```

For a circle point A , the point C , and the antipodal continuation $C - K - X$, it produces an interior ray KT in the exterior-central angle AKX such that

$$\angle AKT \cong \angle XKT$$

and

$$\angle ACK \cong \angle AKT.$$

The same construction applied to B produces a second half-angle ray KU . These two witnessed half-angle configurations are the synthetic replacements for Forder's formulas

$$\angle AKX = 2\angle ACK, \quad \angle BKX = 2\angle BCK.$$

This reusable theorem belongs in `HilbertInterfaceIV`, not in `Proposition4_12.lean`, because later circle arguments can use the same exterior-central geometry independently of IV.12.

9 The Addition Branch

Assume now that A and B are on opposite sides of the axis KX . The axis therefore crosses the interior of the central angle. The proof obtains two local bisectors:

$$\begin{aligned} KT &\text{ bisects } \angle AKX, \\ KU &\text{ bisects } \angle XKB. \end{aligned}$$

The four angular components occur in the order

$$\alpha, \alpha, \beta, \beta.$$

The central combinatorial theorem of the branch is the synthetic rearrangement

$$\alpha, \alpha, \beta, \beta \longrightarrow \alpha, \beta \mid \alpha, \beta.$$

In production this is supplied by the reusable theorem

```
hilbert_angleDecomposition_double_sum_bisector
```

A, B on opposite sides of axis KX

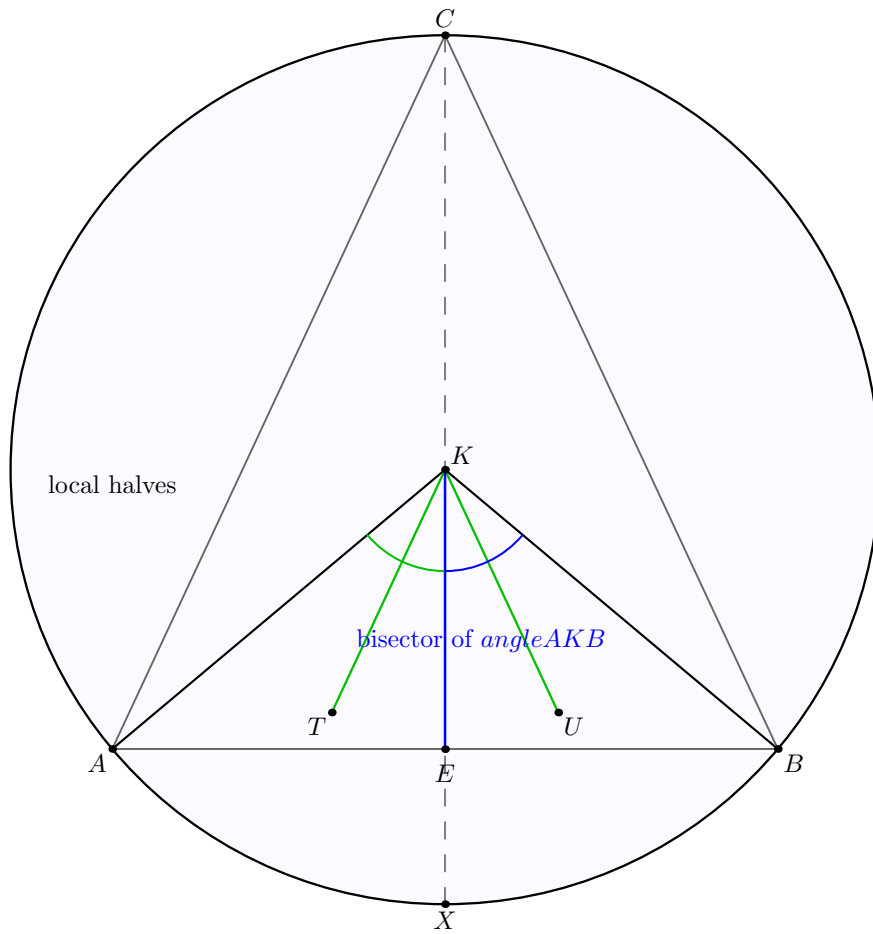


Figure 3: Addition branch of IV.12. The points A and B lie on opposite sides of the axis KX . Two local half-angle constructions are combined into a bisector of the whole central angle.

It constructs a new ray KE inside $\angle AKB$ and proves

$$\angle AKE \cong \angle BKE.$$

At the same time the nested-ray data identify the relevant component of this new half with the corresponding component of the inscribed angle at C . Angle addition then gives

$$\angle ACB \cong \angle AKE.$$

The proposition-specific result is

```
hilbert_forder_IV12_addition_case
```

This branch is a good example of why the formal proof should not be described as “angle arithmetic”. No numerical sum of angle values exists in the proof object. The whole calculation is expressed by interior-ray witnesses and congruence transport.

10 The Subtraction Branch

The same-side branch is structurally more delicate. Assume that A and B are both off the axis KX and lie on the same side of it.

At K , one of the two radius rays lies inside the angle formed by the other radius ray and KX . At C , an analogous nesting occurs among the chord rays and CK . A priori the two choices could be crossed: the “larger” orientation at the centre need not automatically be synchronized with the “larger” orientation at the circumference.

The proof therefore first establishes a ray-order classification and then a synchronization theorem:

```
hilbert_forder_IV12_sameSide_ray_orders
hilbert_forder_IV12_sameSide_orders_synchronized
```

The synchronized alternatives are exactly:

KA is interior to $\angle BKK$ and CA is interior to $\angle BCK$,
or
 KB is interior to $\angle AKK$ and CB is interior to $\angle ACK$.

The synchronization step uses strict monotonicity of angle halves: crossed orientations would force incompatible strict comparisons between congruent half-angles.

After synchronization, the proof chooses an ordinary bisector KE of the target central angle $\angle AKB$. It also has the two exterior-central half constructions described above. In either orientation, the already proved addition theorem is used to reconstruct a bisector KW of the larger exterior-central angle from

$$\text{half}(\text{target}) + \text{half}(\text{smaller exterior angle}).$$

Uniqueness of angle halves identifies this reconstructed bisector with the previously constructed exterior-central bisector. The two complete angle decompositions now have matching larger parts

A, B on the same side of axis KX

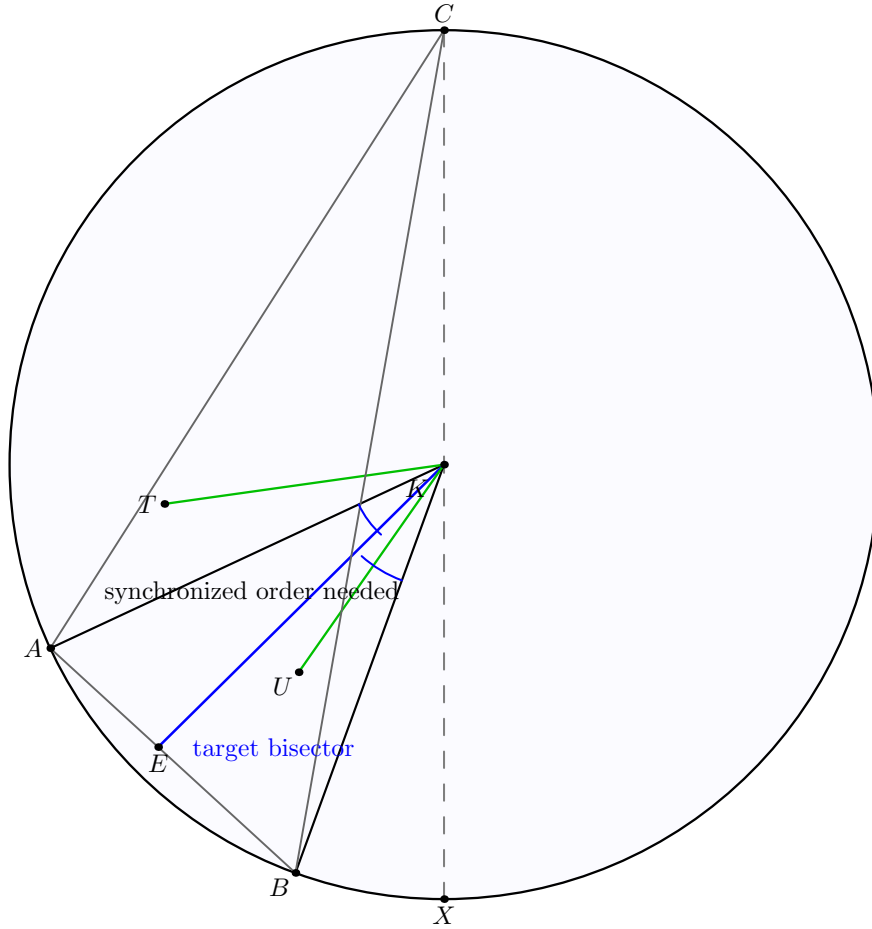


Figure 4: Subtraction branch of IV.12. The points A and B lie on the same side of the axis KX . The proof requires an additional synchronization theorem for the ray order before the final synthetic subtraction step can be carried out.

and matching smaller parts. A single synthetic subtraction principle compares the remaining components, giving

$$\angle ACB \cong \angle AKE.$$

The production helper closing this branch is

```
hilbert_forder_IV12_subtraction_case
```

This is the formal counterpart of Forder’s line

$$2\beta - 2\alpha = 2(\beta - \alpha),$$

but no subtraction operation on numerical angle measures is introduced.

11 Completion of IV.12

Once the four branches have been isolated, the public proof is conceptually short.

1. Construct the antipodal point X with $C - K - X$.
2. Let KX be the reference axis.
3. If A lies on the axis, use the first diameter branch.
4. Else, if B lies on the axis, use the symmetric diameter branch.
5. Otherwise both points are off the axis.
6. If they lie on the same side, use the subtraction branch.
7. If they do not lie on the same side, plane separation gives opposite sides, and the addition branch applies.

This final case split is implemented by the public theorem

```
hilbert_IV12_central_half
```

after the production refactor. The earlier development-stage theorem performed exactly the same exhaustive classification; the production file retains the mathematics but removes stage-labelled public API.

12 Forder IV.12.1: Equal Angles in the Same Segment

Forder immediately records the same-segment corollary: if A, B, C, D lie on one circle and C, D, K lie in the same half-plane determined by chord AB , then

$$\angle ACB \cong \angle ADB.$$

The formal proposition interface is

```

def HilbertForderIV12_1SameSegment
  [H : HilbertIncidence Geo]
  [_HO : @HilbertOrder Geo H] : Prop :=
  forall K R A B C D : Geo.Point,
    forall chord : Geo.Line,
      Ne A B ->
        H.OnLine A chord ->
        H.OnLine B chord ->
        HilbertCircle Geo K R A ->
        HilbertCircle Geo K R B ->
        HilbertCircle Geo K R C ->
        HilbertCircle Geo K R D ->
        HilbertSameSide Geo C K chord ->
        HilbertSameSide Geo D K chord ->
        Geo.AngleCongruent A C B A D B

```

The proof is intentionally short once IV.12 is available. Apply IV.12 to C and to D . Both inscribed angles are represented as halves of the same central angle AKB . The synthetic analogue of Forder IV.46.2 then identifies those halves.

In production this reduction is isolated as

```

hilbert_forder_IV12_1_of_IV12_aux

```

and the closed public theorem is

```

theorem hilbert_IV12_same_segment
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV12_1SameSegment (Geo := Geo)

```

Thus IV.12.1 is not reproved by reopening the full circle configuration. It is a genuine corollary of the half-central-angle theorem.

13 Lean Formalization

At the highest level the production proof has a small public surface and a substantial proposition-specific interior.

The public endpoints are:

```

hilbert_IV12_central_half
hilbert_IV12_same_segment

```

The proposition-specific proof architecture is represented by:

```

hilbert_forder_IV12_1_of_IV12_aux
hilbert_forder_IV12_addition_configuration
hilbert_forder_IV12_addition_case

```

```

hilbert_forder_IV12_sameSide_ray_orders
hilbert_forder_IV12_sameSide_orders_synchronized
hilbert_forder_IV12_subtraction_case
hilbert_forder_IV12_diameter_A
hilbert_forder_IV12_diameter_B

```

These declarations remain local to the numbered proposition because their purpose is to organize the proof of IV.12 itself.

By contrast, the following kinds of lemmas were promoted to the reusable interface:

```

hilbert_circle_antipode
hilbert_circle_center_midpoint_of_chord
hilbert_circle_exterior_central_half
hilbert_circle_chord_inner_point_inside
hilbert_circle_chord_axis_order
hilbert_angle_bisector_with_interior
hilbert_angleDecomposition_swap_components
hilbert_angleDecomposition_nested_inside_left
hilbert_angleDecomposition_nested_inside_right
hilbert_angleDecomposition_middle_divider
hilbert_angleDecomposition_double_sum_bisector
hilbert_angleDecomposition_half_less_of_whole_less

```

The distinction is semantic rather than historical. A lemma belongs to `HilbertInterfaceIV` if it states a general circle/order/angle fact that a later proposition may reuse. It belongs to `Proposition4_12.lean` if it exists only to organize the IV.12 case analysis.

14 Hilbert, Book Zero, and Earlier Euclid Components

14.1 Incidence and plane separation

The theorem uses explicit carrier lines, same-side and opposite-side relations, and line uniqueness. These replace the informal visual judgment that a point or ray lies “inside” one of Forder’s diagrams.

The top-level addition/subtraction split is ultimately a plane-separation statement: once A, B are known to be off the axis, failure of same-side implies opposite-side.

14.2 Order and betweenness

The antipodal configuration $C - K - X$, midpoint configurations on diameters, intersection points on chords, and all interior-angle witnesses are encoded by betweenness or by `HilbertRayMeetsSegment`. This order information is not auxiliary bookkeeping: it determines whether angle components are to be added or subtracted.

14.3 Congruence

Equal radii produce isosceles triangles; angle congruence transports the equal base-angle information into the decompositions used by the proof. Congruence also supplies the synthetic notion of equal angle halves.

14.4 Euclid I.32

The exceptional diameter branch invokes the exterior-angle theorem I.32. The current theorem signature therefore uses

```
[HE : @HilbertEuclideanPlane Geo H]
```

rather than only the neutral incidence/order/congruence layers.

14.5 Angle-decomposition infrastructure

The generic addition and subtraction branches depend on the project's synthetic angle-decomposition library. Its role is not to assign magnitudes to angles but to compare and rearrange witnessed interior decompositions. This is the formal replacement for the algebraic appearance of Forder's angle calculations.

15 Foundational Boundary

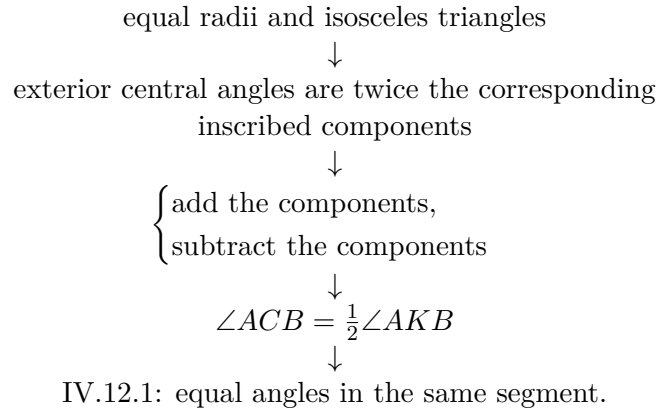
IV.12 introduces no proposition-local geometric axiom. Its public theorem is proved from the existing `HilbertEuclideanPlane` infrastructure and from reusable theorems already available in the production library.

The mathematically important boundary exposed by the proof is not a new continuity principle but the transition from neutral circle/order/congruence reasoning to the Euclidean angle machinery used in the diameter branch and the general angle constructions.

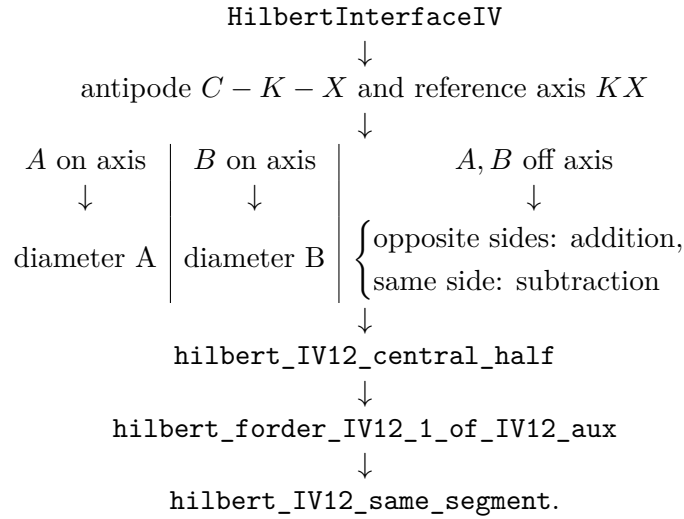
A separate raw `#print axioms` table is intentionally not reproduced here. As in the documentation of the other books, the purpose of this section is to explain the geometric role of the foundational layers rather than to copy implementation output.

16 Classical and Formal Dependency Structures

Forder's conceptual dependency is:



The current production architecture is:



At module level the direct dependency is simply

```
HilbertInterfaceIV
-> Proposition4_12
```

because the reusable support extracted during the proof has been moved out of the numbered module.

17 Diagrammatic Audit

IV.12 is one of the propositions for which figures are mathematically useful, not decorative. A complete documentation pass should contain three Asymptote figures.

1. **Forder source configuration.** Circle with centre K , chord AB , point C , antipode X , and the axis KX . The figure should show the role of the same-side hypothesis for C, K relative to AB .

2. **Addition branch.** A, B on opposite sides of KX , together with the two local half-angle rays and the reconstructed bisector of $\angle AKB$. The purpose is to make the $\alpha, \alpha, \beta, \beta$ rearrangement visible.
3. **Subtraction branch.** A, B on the same side of KX , showing the two synchronized nesting alternatives and the target bisector. This figure should explain why a genuine order theorem is needed before synthetic subtraction can be applied.

The two diameter endpoint cases can be shown as small panels in the first figure rather than as separate full figures.

18 Conclusion

Forder IV.12 is not merely an elementary statement about an inscribed angle being half a central angle. In a strictly synthetic reconstruction it is a theorem about the existence and position of a bisecting ray.

The formal proof exposes exactly the order distinctions suppressed by the classical diagram. The antipodal axis produces two exceptional diameter cases and two generic cases. The opposite-side case is solved by synthetic angle addition; the same-side case requires synchronized ray order and a synthetic subtraction argument. IV.12.1 then becomes a clean corollary: two inscribed angles in the same segment are two halves of the same central angle.

The proposition also fixes an important architectural pattern for the rest of Forder Book IV. General circle and angle-decomposition facts belong to `HilbertInterfaceIV`; only the proposition-specific branching remains inside `Proposition4_12.lean`. The resulting production module is therefore both mathematically transparent and reusable downstream.

References

- [1] Henry George Forder, *Foundations of Euclidean Geometry*, Cambridge University Press, Cambridge, 1927.